1. In *Drosophila*, red eye *(R)* is dominant to orange eye *(r)* and straight wing *(W)* is dominant to curled wing *(w)*. A true-breeding fly with orange eyes and curled wings is crossed with a true-breeding fly having red eyes and straight wings. The F_1 flies are then allowed to mate.

(a) Using the forked line method, list all possible F_2 phenotypes and calculate the probability of each.

(b)The F_2 generation consisted of 406 flies with red eyes and straight wings; 47 flies with orange eyes and curled wings; 155 flies with red eyes and curled wings; and 173 flies with orange eyes and straight wings. Using the chi- square test, determine if this outcome is consistent with the predicted ratios for this cross.

1. (a) true-breeding fly with orange eyes and curled wings: wwrr

true-breeding fly with red eyes and straight wings: WWRR

F_1 flies: WwRr

F_2:

$\frac{3}{4}W$ / $\frac{3}{4}R \longrightarrow \frac{3}{4}\times\frac{3}{4}WR=\frac{9}{16}WR$

$\frac{3}{4}W$ \ $\frac{1}{4}r \longrightarrow \frac{3}{4}\times\frac{1}{4}Wr=\frac{3}{16}Wr$

$\frac{1}{4}w$ / $\frac{3}{4}R \longrightarrow \frac{1}{4}\times\frac{3}{4}wR=\frac{3}{16}wR$

$\frac{1}{4}w$ \ $\frac{1}{4}r \longrightarrow \frac{1}{4}\times\frac{1}{4}wr=\frac{1}{16}wr$

(b) According to the question, there are:

Observed data: 406 WR, 173 Wr, 155 wR, 47 wr

Total number: 781

Expected data:

$WR: 781\times\frac{9}{16}=439.3125$

$Wr: 781\times\frac{3}{16}=146.4375$

$wR: 781\times\frac{3}{16}=146.4375$

$wr: 781\times\frac{1}{16}=48.8125$

$\chi^2=\sum\frac{(O-E)^2}{E}$

$=\frac{(406-439.3125)^2}{439.3125}+\frac{(173-146.4375)^2}{146.4375}+\frac{(155-146.4375)^2}{146.4375}+\frac{(47-48.8125)^2}{48.8125}$

≈ 7.91

df: $n=3$. Refer to a Chi Square Table,

$0.01<P<0.05$, so it's less likely to be consistent.

2. Consider the following crosses in peas. The two traits being investigated involve the shape and the color of flowers. The outcomes of four crosses are shown below.

Parents of cross	Number of offspring of each phenotype			
	White, axillary	White, terminal	Yellow, axillary	Yellow, terminal
Yellow, terminal Yellow, axillary	0	0	63	71
Yellow, axillary Yellow, axillary	29	11	95	33
White, axillary White, axillary	85	26	0	0
Yellow, axillary Yellow, terminal	20	22	68	63

(a) Which color is dominant and which is recessive?

(b) Which flower shape is dominant and which is recessive?

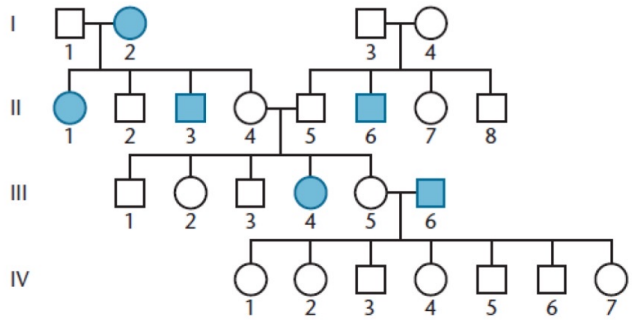
(c) What is the most likely genotype of each parent for each of the four crosses?

(a) Yellow is dominant,
white is recessive.

(b) Axilliary is dominant,
terminal is recessive.

(c) Assume: Yellow (Y), white (y), axilliary (A), terminal (a).
group ①: YYaa × YYAa. 或 Yyaa × YYAa. 或 YYaa × YyAa.
group ②: YyAa × YyAa.
group ③: yyAa × yyAa.
group ④: YyAa × Yyaa.

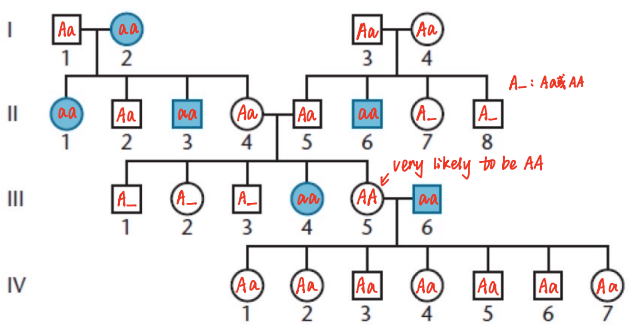
3. Consider the following pedigree.



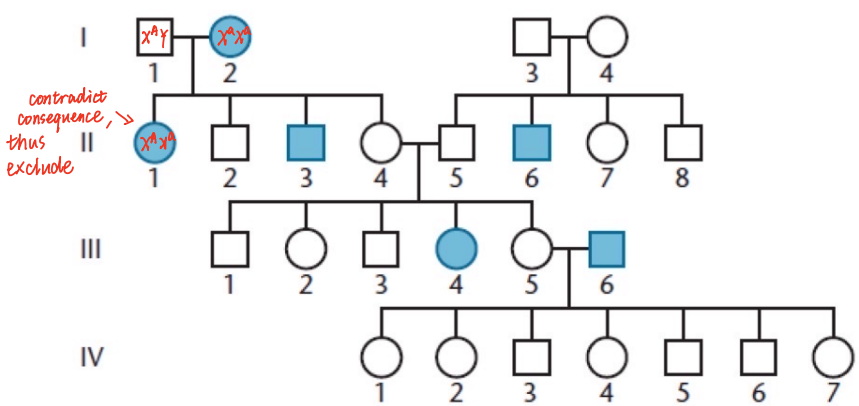
Predict the mode of inheritance of the trait of interest and the most probable genotype of each individual. Assume that the alleles *A* and *a* control the expression.

It's obvious that the virulence gene is recessive.

Assume that it's autosomal recessive inheritance. (a)



Assume that it's with X chromosome recessive inheritance. (X^a)



And its obvious that it is not with Y chromosome recessive inheritance.
From what I discussed above, the mode of inheritance is autosomal recessive inheritance.